Hopfield Network Exercise (Answers)

Neural Computing

March 4, 2026

1 Pattern Storage & Weight Matrix Calculation

Consider a Hopfield network with 9 neurons arranged in a 3×3 grid. We will use bipolar representation where each neuron can either be +1 or -1.

We want to store the following two patterns:

- Pattern A: A horizontal line

  $$\begin{bmatrix} +1 & +1 & +1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

- Pattern B: A vertical line

  $$\begin{bmatrix} +1 & -1 & -1 \\ +1 & -1 & -1 \\ +1 & -1 & -1 \end{bmatrix}$$

Questions:

1. Convert patterns A and B into vector form, where each vector has 9 elements.

   $\vec{\xi^A}$:

$$\begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

$\vec{\xi^B}$:

$$\begin{bmatrix} +1 \\ -1 \\ -1 \\ +1 \\ -1 \\ -1 \\ +1 \\ -1 \\ -1 \end{bmatrix}$$

2. Using the Hebbian learning rule, calculate the weight matrix W. Recall that:

   - $w_{ij} = \frac{1}{N} \sum_{\mu}^{P} \xi_i^{\mu} \xi_j^{\mu}$ for all patterns μ.
   - The diagonal elements w_{ii} should be set to 0 (no self connections).
   - Show your calculations for at least 3 specific weights.

   (We show the calculation for one weight)

$$\begin{aligned} w_{1,5} &= \frac{1}{9} \left[(\xi_1^A \xi_5^A) + (\xi_1^B \xi_5^B) \right] \\ &= \frac{1}{9} [(1 \cdot -1) + (1 \cdot -1)] \\ &= -\frac{2}{9} \end{aligned}$$

2 Pattern Retrieval

Questions:

3. Consider a new input pattern C

$$\begin{bmatrix} +1 & +1 & +1 \\ -1 & -1 & -1 \\ +1 & -1 & -1 \end{bmatrix}$$

This pattern has one neuron different from Pattern A. Now:

(a) Convert pattern C to vector form

$\vec{x^C}$:

$$\begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \\ +1 \\ -1 \\ -1 \end{bmatrix}$$

(b) Calculate the initial state of each neuron after one synchronous update. Show your calculations for at least 3 specific weights. Recall that the update rule is:

$$O_i(t+1) = \mathrm{sgn}\left(\sum_j^N w_{ij} O_j(t)\right)$$

(After the weights of our matrix W were initialized to store patterns A and B, we input vector C to the network)

$$\begin{aligned} \vec{x}^A \cdot \vec{x}^C &= (1 \cdot 1) + (1 \cdot 1) + (1 \cdot 1) + ((-1) \cdot (-1)) + ((-1) \cdot (-1)) + \\ &\quad ((-1) \cdot (-1)) + ((-1) \cdot 1) + ((-1) \cdot (-1)) + ((-1) \cdot (-1)) \\ &= 7 \end{aligned}$$

$$\vec{x}^B \cdot \vec{x}^C = 3$$

For neuron 7,

$$
\begin{aligned}
O_7(1) = \sum_i^N w_{ij} O_j(0) &= \sum_i^N \left(\frac{1}{N} \sum_\mu^p \xi_i^\mu \xi_j^\mu \right) O_j(0) \\
&= \frac{1}{N} \sum_i^N \sum_\mu^0 \xi_i^\mu \xi_j^\mu x_j \\
&= \frac{1}{N} \sum_i^N \xi_i^\mu \sum_\mu^0 \xi_j^\mu x_j \\
&= \frac{1}{N} \sum_i^N \xi_i^\mu (\xi^\mu \cdot x(t))
\end{aligned}
$$

$$
\begin{aligned}
O_7(1) &= \operatorname{sgn}\left(\frac{1}{N} \sum \xi_7^\mu (\xi^\mu \cdot x(t)) \right), \quad \mathrm{N} = 9 \\
&= \operatorname{sgn}\left(\frac{1}{9} \left(x_7^A (x^A \cdot x^C) + x_7^B (x^B \cdot x^C) \right) \right) \\
&= \operatorname{sgn}\left(\frac{1}{9} \left((-1) \cdot 7 + 1 \cdot 3 \right) \right) \\
&= \operatorname{sgn}\left(-\frac{4}{9} \right) \\
&= -1
\end{aligned}
$$

The updated state is:

$$
\vec{O}(1) = \begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}
$$

(c) Does the network converge to one of the stored patterns after this update? If not, continue the updates until convergence or until you can explain why it won't converge.

The network converges to pattern A after one update.

3 Energy Function

The energy function of a Hopfield network is given by:

$$E = -\frac{1}{2}\sum_{i}^{N}\sum_{j}^{N} w_{ij}x_i x_j = -\frac{1}{2}\vec{x}^T W \vec{x}$$

Questions:

4. Calculate the energy of Pattern A ($\vec{\xi^A}$), Pattern B ($\vec{\xi^B}$), Pattern C ($\vec{\xi^C}$).
   (We show only the calculations for one energy function.)

$$w_{ij} = \frac{1}{N}\sum_{\mu}^{P} \xi_i^\mu \xi_j^\mu$$

$$W = \frac{1}{N}\sum_{\mu}^{P} \vec{\xi}^\mu (\vec{\xi}^\mu)^T$$

Since only patterns A and B are stored, this becomes

$$W = \frac{1}{N}\left(\vec{\xi}^A(\vec{\xi}^A)^T + \vec{\xi}^B(\vec{\xi}^B)^T\right).$$

$$W = \frac{1}{9}\left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}\right.$$

$$\left. + \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ -1 \\ -1 \\ 1 \\ -1 \\ -1 \end{bmatrix}\begin{bmatrix} 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 \end{bmatrix}\right].$$

(Note: The diagonal elements W_{ii} should be set to 0, no self connections.)

$$
= \frac{1}{9}\left(\begin{bmatrix}
{\color{red}0} & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 \\
1 & {\color{red}0} & 1 & -1 & -1 & -1 & -1 & -1 & -1 \\
1 & 1 & {\color{red}0} & -1 & -1 & -1 & -1 & -1 & -1 \\
-1 & -1 & -1 & {\color{red}0} & 1 & 1 & 1 & 1 & 1 \\
-1 & -1 & -1 & 1 & {\color{red}0} & 1 & 1 & 1 & 1 \\
-1 & -1 & -1 & 1 & 1 & {\color{red}0} & 1 & 1 & 1 \\
-1 & -1 & -1 & 1 & 1 & 1 & {\color{red}0} & 1 & 1 \\
-1 & -1 & -1 & 1 & 1 & 1 & 1 & {\color{red}0} & 1 \\
-1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & {\color{red}0}
\end{bmatrix}\right.
$$

$$
\left. + \begin{bmatrix}
{\color{red}0} & -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 \\
-1 & {\color{red}0} & 1 & -1 & 1 & 1 & -1 & 1 & 1 \\
-1 & 1 & {\color{red}0} & -1 & 1 & 1 & -1 & 1 & 1 \\
1 & -1 & -1 & {\color{red}0} & -1 & -1 & 1 & -1 & -1 \\
-1 & 1 & 1 & -1 & {\color{red}0} & 1 & -1 & 1 & 1 \\
-1 & 1 & 1 & -1 & 1 & {\color{red}0} & -1 & 1 & 1 \\
1 & -1 & -1 & 1 & -1 & -1 & {\color{red}0} & -1 & -1 \\
-1 & 1 & 1 & -1 & 1 & 1 & -1 & {\color{red}0} & 1 \\
-1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & {\color{red}0}
\end{bmatrix}\right)
$$

$$
= \frac{1}{9}\begin{bmatrix}
0 & 0 & 0 & 0 & -2 & -2 & 0 & -2 & -2 \\
0 & 0 & 2 & -2 & 0 & 0 & -2 & 0 & 0 \\
0 & 2 & 0 & -2 & 0 & 0 & -2 & 0 & 0 \\
0 & -2 & -2 & 0 & 0 & 0 & 2 & 0 & 0 \\
-2 & 0 & 0 & 0 & 0 & 2 & 0 & 2 & 2 \\
-2 & 0 & 0 & 0 & 2 & 0 & 0 & 2 & 2 \\
0 & -2 & -2 & 2 & 0 & 0 & 0 & 0 & 0 \\
-2 & 0 & 0 & 0 & 2 & 2 & 0 & 0 & 2 \\
-2 & 0 & 0 & 0 & 2 & 2 & 0 & 2 & 0
\end{bmatrix}
$$

$$
= \frac{2}{9}\begin{bmatrix}
0 & 0 & 0 & 0 & -1 & -1 & 0 & -1 & -1 \\
0 & 0 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\
0 & 1 & 0 & -1 & 0 & 0 & -1 & 0 & 0 \\
0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\
-1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\
0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\
-1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0
\end{bmatrix}
$$

$$E_A = -\frac{1}{2}\cdot\frac{2}{9}\begin{bmatrix}+1 & +1 & +1 & -1 & -1 & -1 & -1 & -1 & -1\end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 & -1 & -1 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

$$= -\frac{1}{9}(4+3+3+3+4+4+3+4+4) = -\frac{32}{9} = -3.56$$

$E_B = -3.56$, A and B are stored symmetrically $\rightarrow$ same attractor depth $\rightarrow$ same energy.

$$E_C = -\frac{1}{2}\cdot\frac{2}{9}\begin{bmatrix}+1 & +1 & +1 & -1 & -1 & -1 & +1 & -1 & -1\end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 & -1 & -1 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \\ +1 \\ -1 \\ -1 \end{bmatrix}$$

$$= -\frac{1}{9}(4+1+1+1+4+4-3+4+4) = -\frac{20}{9} = -2.22$$

- Pattern A: -3.56
- Pattern B: -3.56
- Pattern C: -2.22

5. Based on your energy calculations, explain:

   (a) Why the stored patterns have lower energy values

   The stored patterns A and B both have an energy of -3.56, while pattern C has -2.22. The stored patterns have lower (more negative) *energy values because the weight matrix was specifically constructed using the Hebbian learning rule to create energy minima at those patterns.*

(b) How the network uses energy minimization to perform pattern completion

The Hopfield network dynamics always move toward states of lower energy. When given a partial or corrupted pattern (like Pattern C, which is just one neuron away from Pattern A), the network updates each neuron to minimize the energy function. In our example, Pattern C had an energy of -2.22, whereas Pattern A had -3.56. By flipping the single different neuron (neuron 7), the network moved from Pattern C to Pattern A, reducing the energy from -2.22 to -3.56.